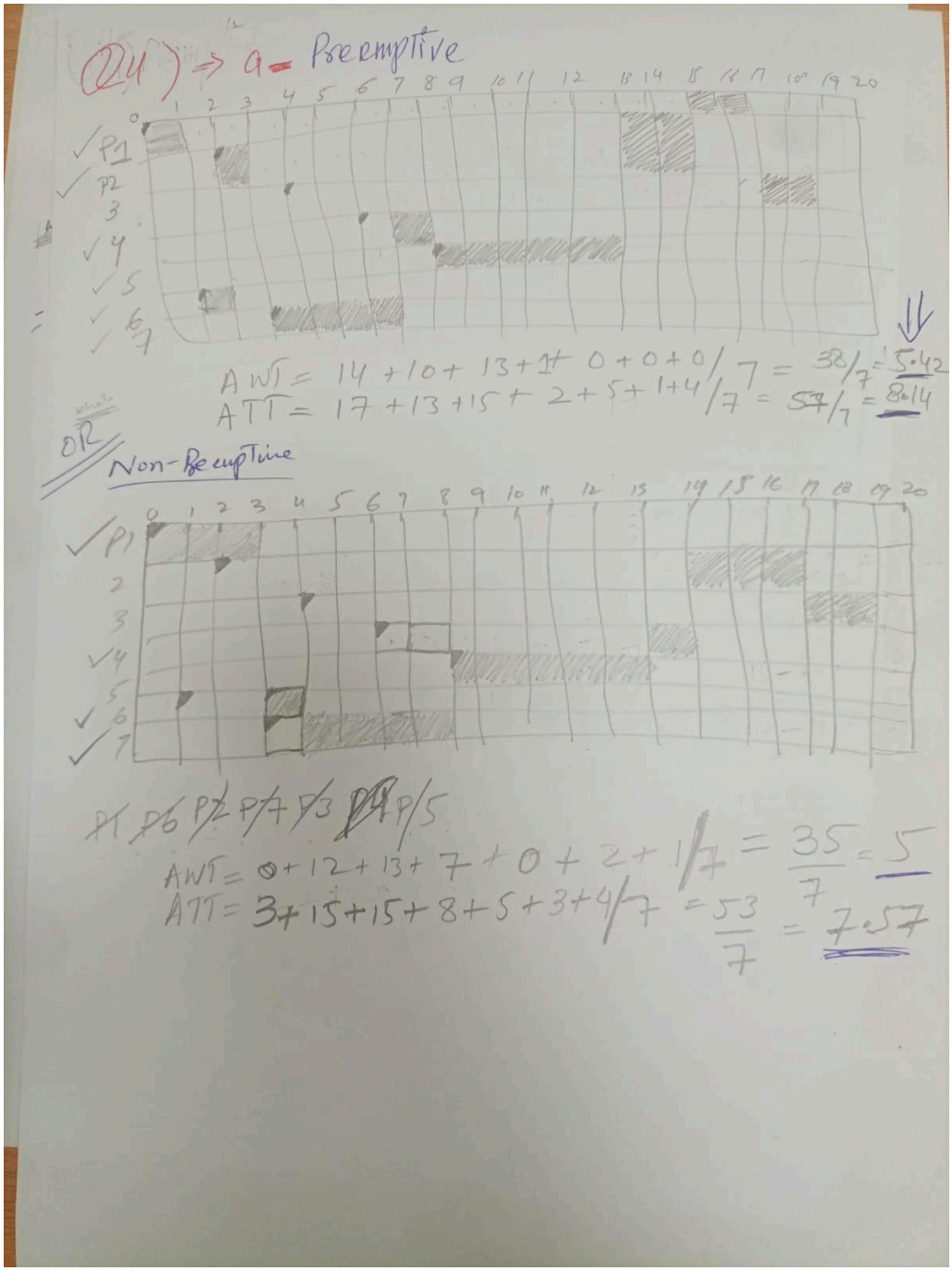
Question 4 [14 Marks]

Considering the table of process provided below, answer the following questions:

Process	Arrival Time	Initial Priority (0 = lowest)	Burst Time
P1	0	1	3
P2	2	2	3
P3	4	1	2
P4	6	4	1
P5	8	5	5
P6	1	3	1
P7	3	4	4

a. What would be the Average Waiting Time and Average Turnaround Time if Priority Scheduling is applied. [1+1+2 Marks]

A.W.T = (14) + (10) + (13) + (1) + (0) + (0) + (0) /7 = 5.42
A.T.T = (17) + (13) + (15) + (2) + (5) + (1) + (4) / 7 = 8.14
Gantt Chart

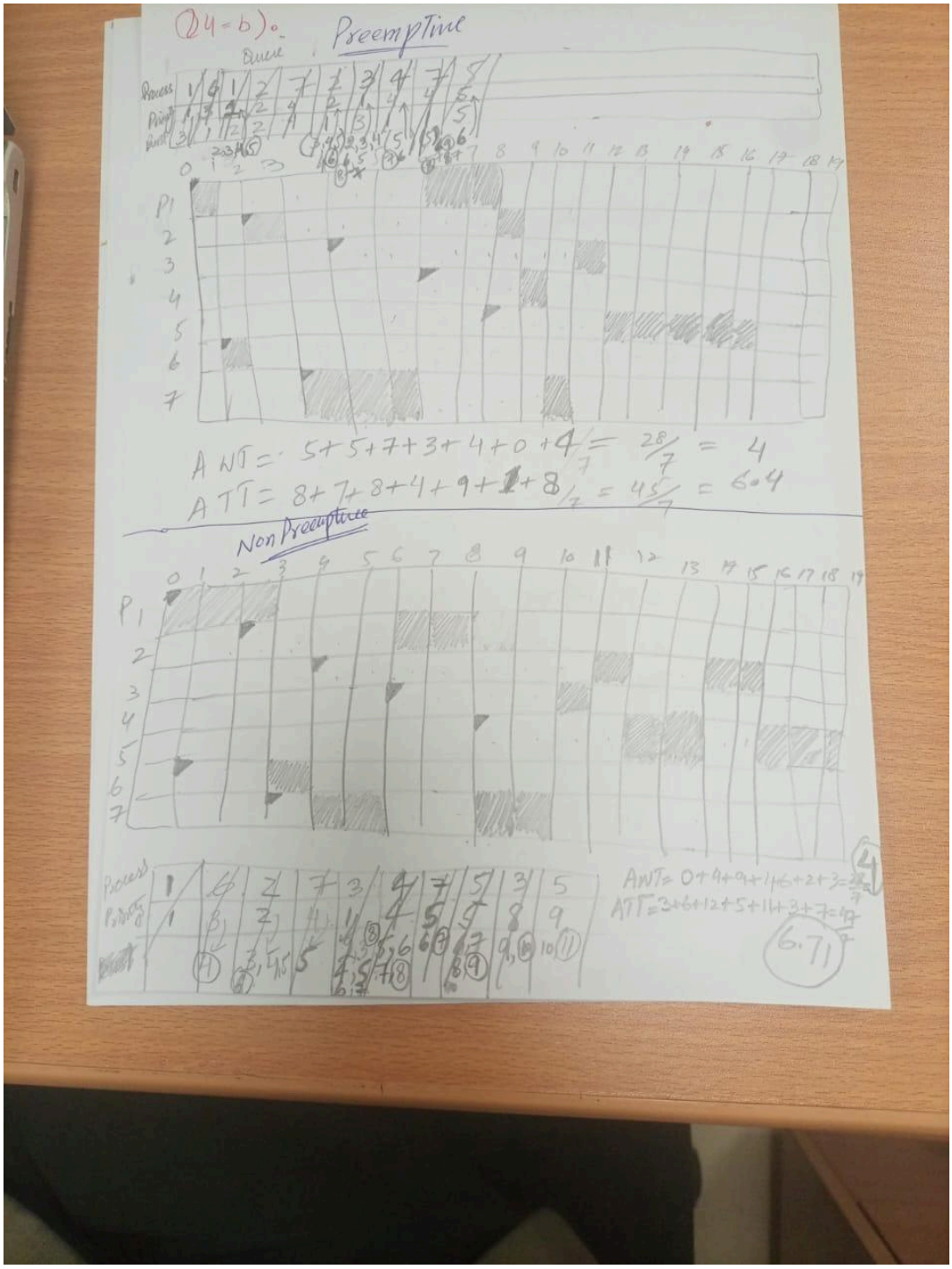


b. What would be the Average Waiting Time and Average Throughput time if Priority Scheduling with aging is applied. For aging you can say that a priority of a waiting process is incremented after 1 unit of execution. [1+1+2 Marks]

A.W.T = ( ) + ( ) + ( ) + ( ) + ( ) + ( ) + ( ) = _____

A.T.T = ( ) + ( ) + ( ) + ( ) + ( ) + ( ) + ( ) = _____

Gantt Chart:



Suppose that all processes have equal priority in the table provided above, which scheduling strategy would be suitable in this case.[2]

Round Robin or Priority Based Round Robin

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Justification:

Round Robin (RR) or Priority-based Round Robin (P-RR) might be suitable. RR ensures each process gets an equal share of the CPU, while P-RR adds a priority consideration.

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c. Consider the following example, IO Start point (relevant to process) and IO duration is also provided for each process.
For Example: P1 is starting IO 1 unit after getting the CPU and IO takes 5 units of time. P3 starts IO immediately after it gets the CPU and takes 3 units for IO.

Process	Arrival Time	Initial Priority (0 = lowest)	IO Start - Duration	CPU Burst Time
P1	0	1	1 - 5	3
P2	2	2	1 - 2	3
P3	4	1	0 - 3	2
P4	6	4	1 - 1	1
P5	8	5	0 - 0	5
P6	1	3	0 - 0	1
P7	3	4	1 - 1	4

We know that when a process is doing its IO it is placed in a blocked queue. Suppose there is no more assumption on IO and IO completes in the exact mentioned duration.

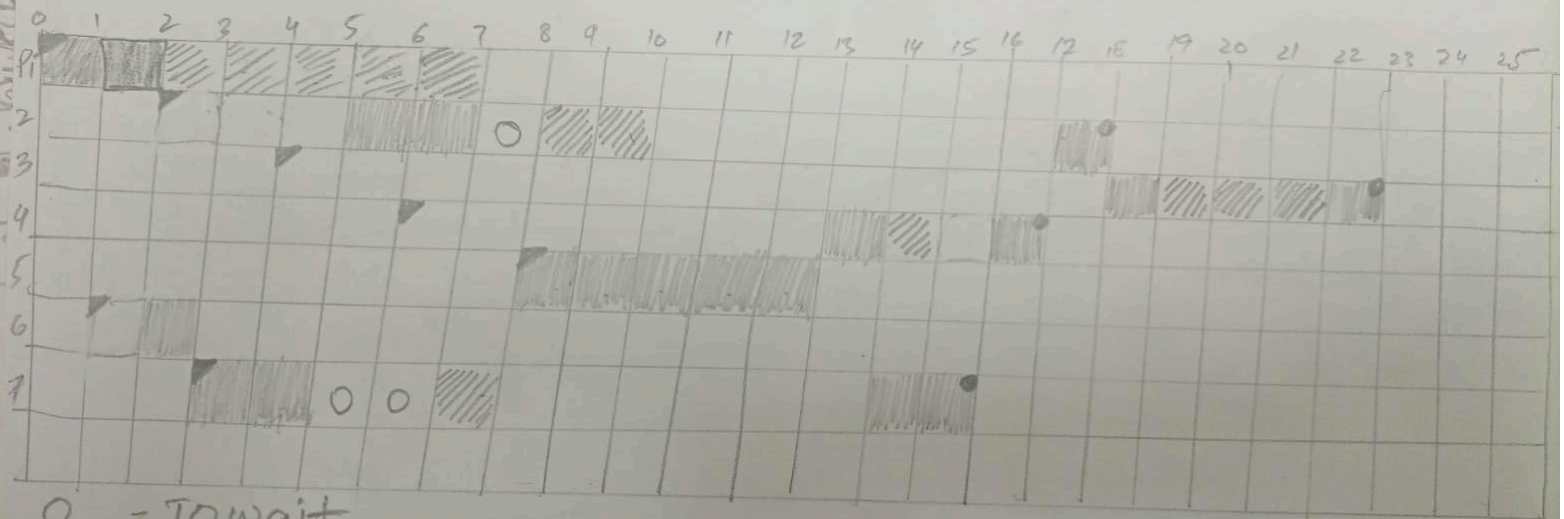
Draw the gantt chart for Priority Based Scheduling again, considering the IO. Also, show the ready and blocked queue states. [4]

Ready Queue

P1	P6	P2	P7	P3	P4	P7	P5	P2	P4
1(3)	3(0)	2(3)	4(4)	1(2)	4(1)	4(2)	5(5)	2(1)	4(0)

Blocked Queue

P1	P7	P2	P4	P3
1(2)	4(2)	2(1)	4(0)	1(1)



O = IO wait

= IO

= CPU

= CPU wait

Question 2

Five processes A, B, C, D and E arrive in this order at the given time with the following CPU bursts. The process is immediately eligible for scheduling i.e. if a process arrives at 1 ms, it can be scheduled at 1 ms.

Process	CPU Burst (ms)	Arrival time (ms)	Priority (1 is lowest)
A	4	6	1
B	7	3	3
C	3	2	2
D	3	3	2
E	5	4	1
F	9	0	3

a). Fill in the entries of the following table with **total** turnaround time and **total** waiting time for each indicated scheduling policy *(**Note**: You do not have to calculate the **average** turnaround / waiting time).* Ignore context switching overhead. **All steps depicting the calculations are mandatory. Filling the table (even with correct entries) without correct solving will lead to zero marks in the question. Only correct entries in the table after proper calculations will secure your marks. No marks will be given for the direct answers. [18 marks]**

Algorithm	**Total** turnaround time (ms)	**Total** waiting Time (ms)
Shortest Remaining Time First	79	48
Round Robin (quantum = 2)	129	98
Priority (Preemptive)	106	75

Show calculations for the above question in the space below (MANDATORY):

Shortest Remaining Time First:

F	F	C	C	C	D	D	D	A	E	F	B

0 1 2 3 4 5 6 7 8 12 17 24 31.

Total TAT = ((8-6)+(12-8)) + ((24-3)+(31-24)) + (5-2) ((8-5)+(5-3)) + ((12-3)+(17-12)) + ((24-17)+(17-0))

= 79.

Total W.T = (8-6) + (24-3) + 0 + (5-3) + (12-4) + (17-2)

= 48.

Round Robin (quantum = 2):

Running queue:

F	C	F	B	D	E	C	A	F	B	D	E	A	F	B	E	F	B

0 2 4 6 8 10 12 13 15 17 19 20 22 24 26 28 29 30 31

Ready Queue

C	F	B	D	E	C	A	F	B	D	E	A	F	B	E	F	B

Total TAT = 18 + 28 + 11 + 17 + 25 + 30

= 129

W.T = 14 + 21 + 8 + 14 + 20 + 21

= 98

Priority (Preemptive):

(c)

F	B	C	D	E	A

0 9 16 19 22 27 31

TAT = (31-6) + (16-3) + (19-2) + (22-3) + (27-4) + 9

= 106

WT = (27-6) + (9-3) + (16-2) + (19-3) + (22-4) + (0)

= 21 + 6 + 14 + 16 + 18 + 0 = 75

b). Schedule the following processes using a Multi-level ***queue scheduling*** algorithm. Assume **Queue1** has a time *quantum* of *4 ms* and **Queue2** has a time *quantum* of *3 ms*.

Process	Arrival Time	CPU burst time (ms)	Arrival Queue
A	**0**	**10**	**1**
B	**2**	**7**	**2**
C	**4**	**6**	**1**
D	**8**	**5**	**2**
E	**12**	**8**	**2**
F	**13**	**7**	**1**

Keep the following things in mind while answering the questions below: [7 marks]

- Queue1 has higher priority than Queue2.
- The arrival queue for the new process is also provided in the table.
- Show you working in the given space. No marks will be given for the direct answers.
- **Your answer should contain numbers only (i.e. time in terms of ms)**

Question	Answer
What will be the ***response time*** for **Process E**? **[2 marks]**	17
What will be the ***waiting time*** for **Process C**? **[2 marks]**	4
What will be the ***turnaround time*** for **Process F**? **[3 marks]**	10

Show the working for the above question in the space below (MANDATORY):